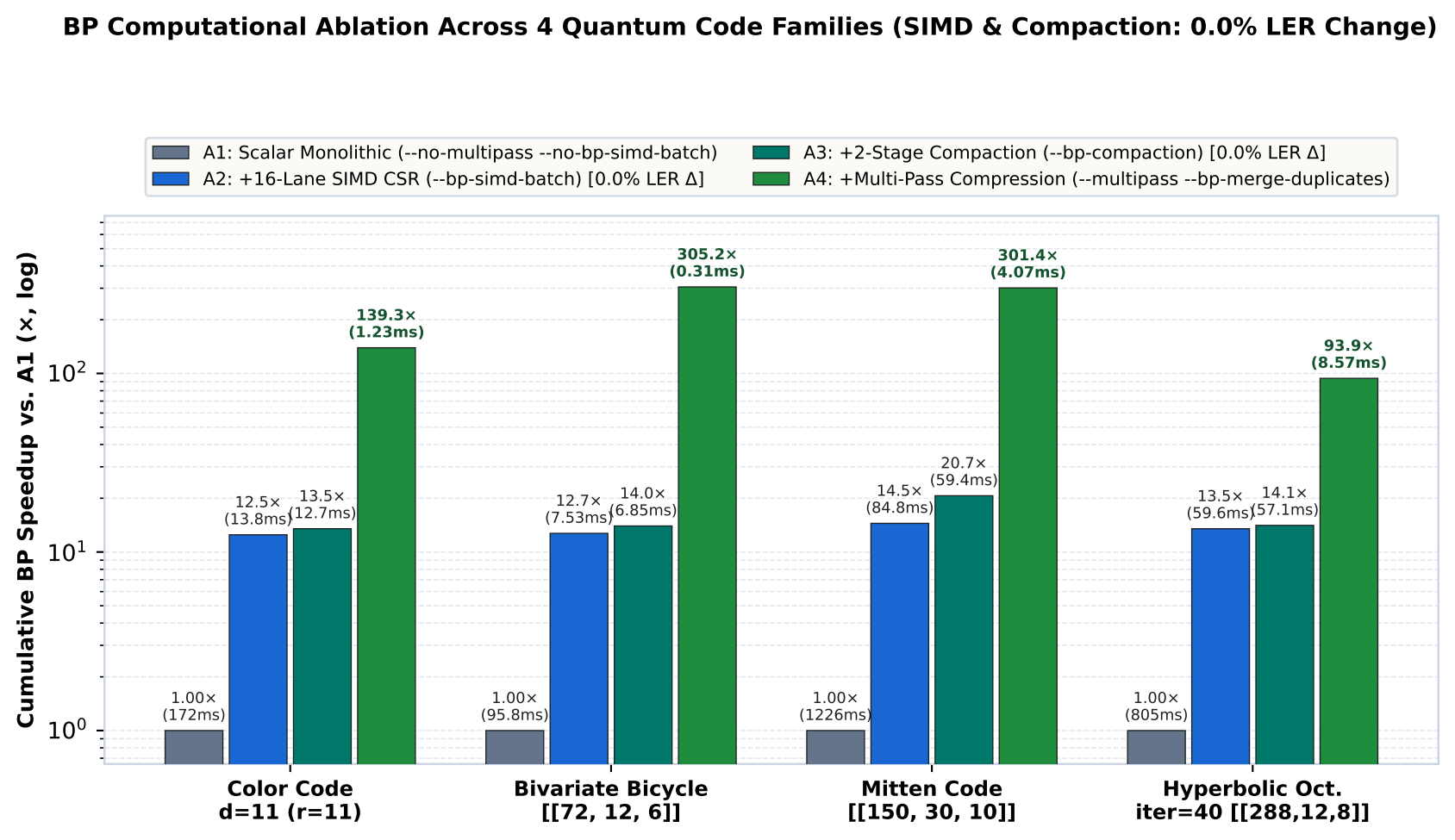
BP Computational Ablation Across 4 Quantum Code Families (SIMD & Compaction: 0.0% LER Change)
A1: Scalar Monolithic (--no-multipass --no-bp-simd-batch)
A2: +16-Lane SIMD CSR (--bp-simd-batch) [0.0% LER Δ]
A3: +2-Stage Compaction (--bp-compaction) [0.0% LER Δ]
A4: +Multi-Pass Compression (--multipass --bp-merge-duplicates)
Cumulative BP Speedup vs. A1 (×, log)
10^0
10^1
10^2
1.00× (172ms)
12.5× (13.8ms)
13.5× (12.7ms)
139.3× (1.23ms)
1.00× (95.8ms)
12.7× (7.53ms)
14.0× (6.85ms)
305.2× (0.31ms)
1.00× (1226ms)
14.5× (84.8ms)
20.7× (59.4ms)
301.4× (4.07ms)
1.00× (805ms)
13.5× (59.6ms)
14.1× (57.1ms)
93.9× (8.57ms)
Color Code d=11 (r=11)
Bivariate Bicycle [[72, 12, 6]]
Mitten Code [[150, 30, 10]]
Hyperbolic Oct. iter=40 [[288,12,8]]